Lab 004 - Normal curve calculations - Solutions.

EPIB607 - Inferential Statistics[a]

[a]Fall 2020, McGill University

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Because the results of random samples include an element of chance, we can't guarantee that our inferences are correct. What we can guarantee is that our methods usually give correct answers. We will see that the reasoning of statistical inference rests on asking "How often would this method give a correct answer if I used it very many times ?" To be able to answer these questions we need to understand sampling distributions and the normal curve. In this exercise you will practice calculating quantiles and probablities from the Normal distribution.

Sampling distribution | Standard error | Normal distribution | Quantiles | Percentiles | Z-scores

1. Normal probability calculations

Using your method of choice, calculate the following probabilities assuming Y is a standard normal distribution ($Y \sim \mathcal{N}(0, 1)$):

a) $P(Y < -1.80)$
b) $P(Y > -1.80)$
c) $P(Y \geq 1.60)$
d) $P(-1.8 < Y \leq 1.6)$

```
# default is always mean = 0, sd = 1, lower.tail=TRUE
# a)
mosaic::xpnorm(q = -1.80)
```

```
#  Registered S3 method overwritten by 'mosaic':
#    method                           from
#    fortify.SpatialPolygonsDataFrame ggplot2
```

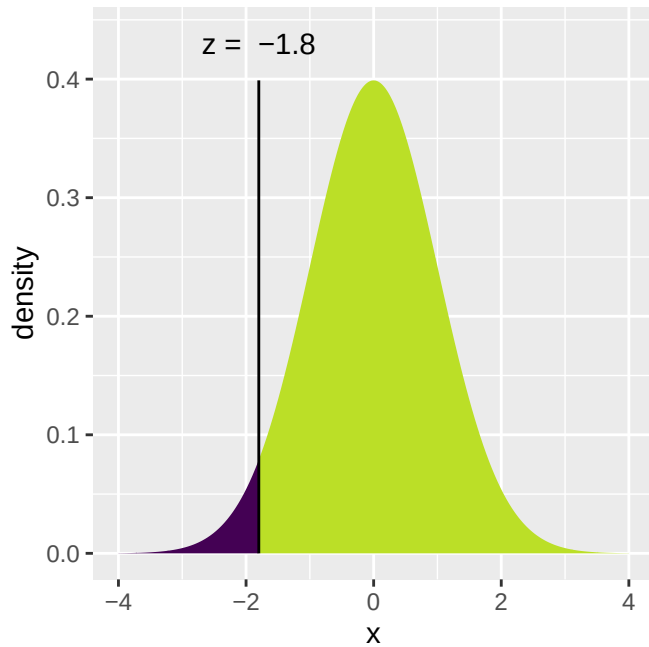
```
#
```

```
#  If X ~ N(0, 1), then
```

```
#   P(X <= -1.8) = P(Z <= -1.8) = 0.03593
```

```
#   P(X >  -1.8) = P(Z >  -1.8) = 0.9641
```

```
#
```




```
#  [1] 0.03593032
```

```
# b)
mosaic::xpnorm(q = -1.80, lower.tail = FALSE)
```

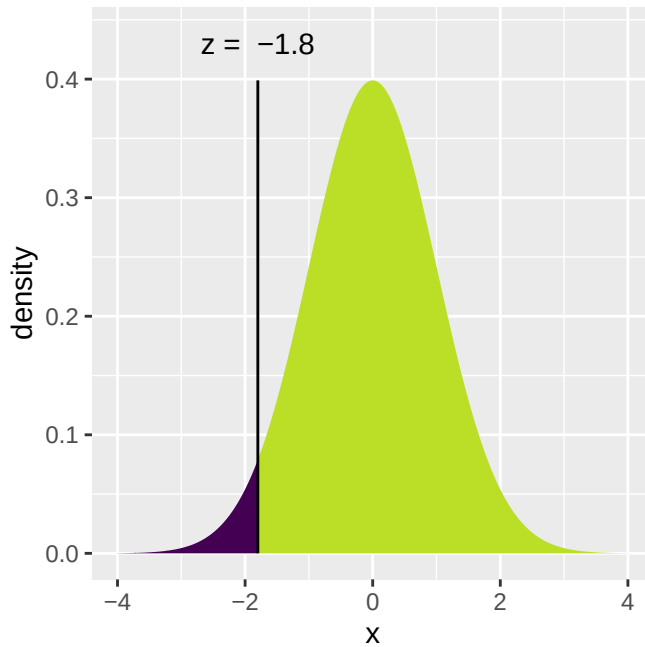
```
#
```

```
#  If X ~ N(0, 1), then
```

```
#   P(X <= -1.8) = P(Z <= -1.8) = 0.03593
```

```
#   P(X >  -1.8) = P(Z >  -1.8) = 0.9641
```

```
#
```




```
#  [1] 0.9640697
```

```
# c)
mosaic::xpnorm(q = 1.60, lower.tail = FALSE)
```

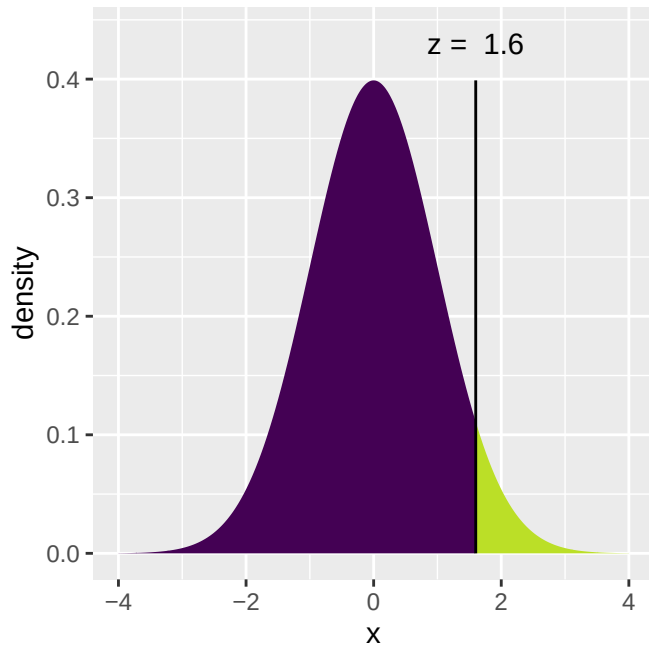
```
#
```

```
#  If X ~ N(0, 1), then
```

```
#   P(X <= 1.6) = P(Z <= 1.6) = 0.9452
```

```
#   P(X >  1.6) = P(Z >  1.6) = 0.0548
```

```
#
```




```
#  [1] 0.05479929
```

```
# d)
pnorm(q = 1.60) - pnorm(q = -1.80)
```

```
#  [1] 0.9092704
```

```
mosaic::xpnorm(q = c(-1.80, 1.60))
```

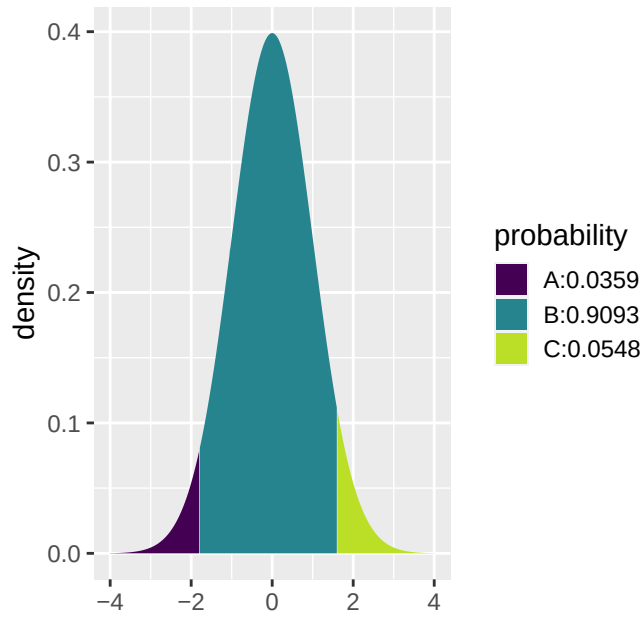
```
#
```

```
#  If X ~ N(0, 1), then
```

```
#   P(X <= -1.8) = P(Z <= -1.8) = 0.03593   P(X <=  1.6) = P(Z <=  1.6) = 0.94520
```

```
#   P(X >  -1.8) = P(Z >  -1.8) = 0.9641    P(X >   1.6) = P(Z >   1.6) = 0.0548
```

```
#
```




```
#  [1] 0.03593032 0.94520071
```

2. HDL cholesterol

a. US women over the age of 19 have a mean (HDL) cholesterol measure of 55 mg/dL with a standard deviation of 15.5 mg/dL. Assume HDL follows a Normal distribution.

   i. What percent of women have low values of HDL, where low is defined to be 40 mg/dL or less?
   ii. HDL levels of 60 mg/dL or more are believed to be protective against heart disease. What percent of women have protective levels of HDL?
   iii. What proportion of women has HDL in the range of 40-60 mg/dL?
   iv. What proportion of women has HDL in the range of 35-65 mg/dL?

```
# i.
mosaic::xpnorm(q = 40, mean = 55, sd = 15.5)
```

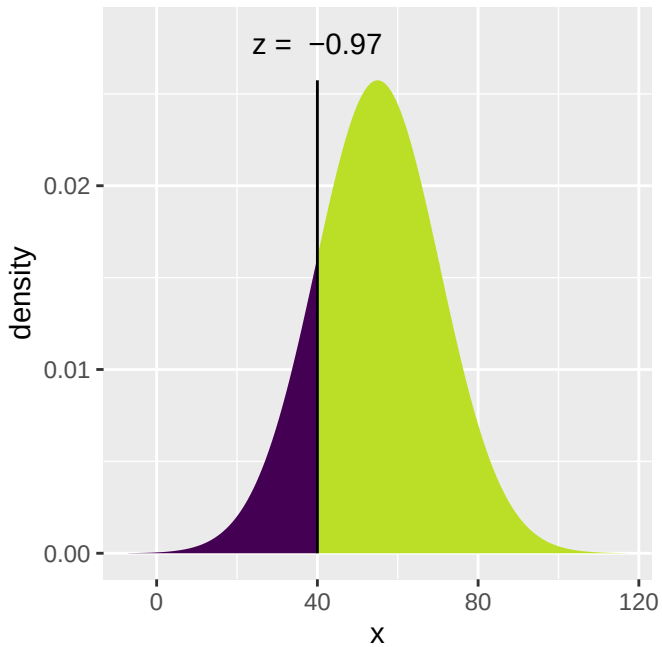
```
#
```

```
#  If X ~ N(55, 15.5), then
```

```
#   P(X <= 40) = P(Z <= -0.9677) = 0.1666
```

```
#   P(X >  40) = P(Z >  -0.9677) = 0.8334
```

```
#
```




```
#  [1] 0.1665866
```

```
# ii.
mosaic::xpnorm(q = 60, mean = 55, sd = 15.5, lower.tail = FALSE)
```

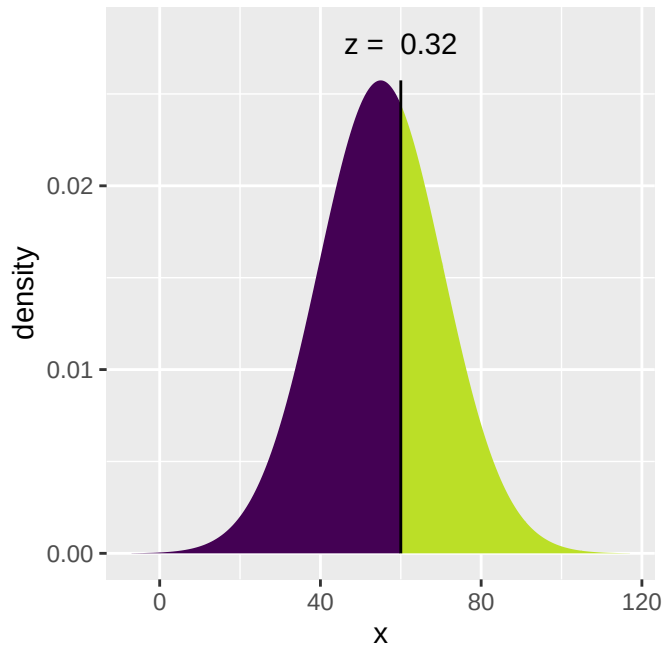
```
#
```

```
#  If X ~ N(55, 15.5), then
```

```
#   P(X <= 60) = P(Z <= 0.3226) = 0.6265
```

```
#   P(X >  60) = P(Z >  0.3226) = 0.3735
```

```
#
```




```
#  [1] 0.3735064
```

```
# iii.
mosaic::xpnorm(q = c(40,60), mean = 55, sd = 15.5)
```

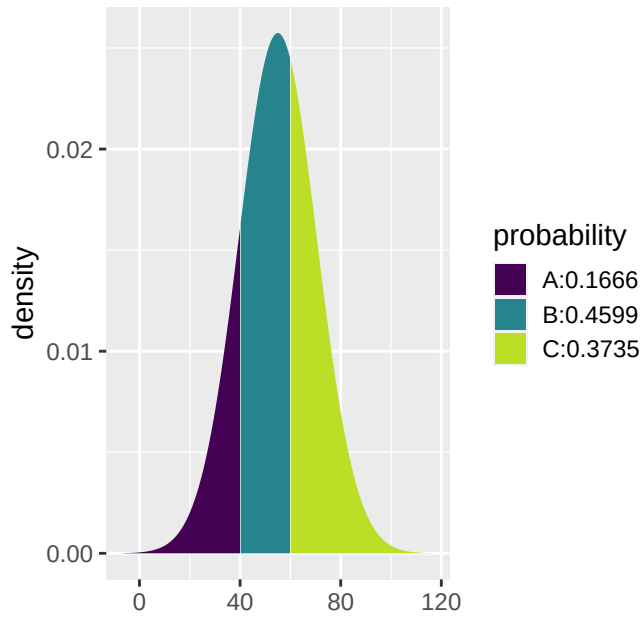
```
#
```

```
#  If X ~ N(55, 15.5), then
```

```
#   P(X <= 40) = P(Z <= -0.9677) = 0.1666   P(X <= 60) = P(Z <=  0.3226) = 0.6265
```

```
#   P(X >  40) = P(Z >  -0.9677) = 0.8334   P(X >  60) = P(Z >   0.3226) = 0.3735
```

```
#
```




```
#  [1] 0.1665866 0.6264936
```

```
pnorm(q = 60, mean = 55, sd = 15.5) - pnorm(q = 40, mean = 55, sd = 15.5)
```

```
#  [1] 0.4599069
```

```
# iv.
mosaic::xpnorm(q = c(35,65), mean = 55, sd = 15.5)
```

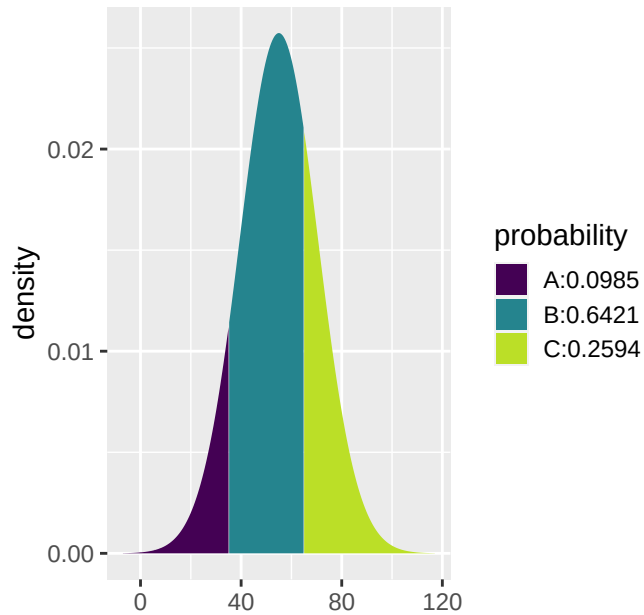
```
#
```

```
#  If X ~ N(55, 15.5), then
```

```
#   P(X <= 35) = P(Z <= -1.2903) = 0.09847  P(X <= 65) = P(Z <=  0.6452) = 0.74059
```

```
#   P(X >  35) = P(Z >  -1.2903) = 0.9015   P(X >  65) = P(Z >   0.6452) = 0.2594
```

```
#
```




```
#  [1] 0.09846934 0.74058867
```

```
pnorm(q = 65, mean = 55, sd = 15.5) - pnorm(q = 35, mean = 55, sd = 15.5)
```

```
#  [1] 0.6421193
```

3. Osteoporosis

Osteoporosis is a condition in which the bones become brittle due to loss of minerals. To diagnose osteoporosis, an elaborate apparatus measures bone mineral density (BMD). BMD is usually reported in standardized form. The standardization is based on a population of healthy young adults. The Wolrd Health Organization (WHO) criterion for osteoporosis is a BMD 2.5 or more standard deviations below the mean for young adults. BMD measurements in a population of people who are similar in age and sex roughly follow a standard normal distribution.

a. What percent of healthy young adults have osteoporosis?
b. Woman aged 70 to 79 are, of course, not young adults. The mean BMD in this age is about -2 on the standard scale for young adults. Suppose that the standard deviation is the same for young adults. What percent of this older population has osteoporosis?
c. Likewise, osteopenia is low BMD, defined by the WHO as a BMD between 1 and 2.5 standard deviations below the mean of young adults. What percent of healthy young adults have osteopenia?
d. The mean BMD among women aged 70 to 79 is about -2 on the standard scale for young adults. Suppose that the standard deviation is the same as for young adults. What percent of this older population has osteopenia?

```r
# a)
mosaic::xpnorm(q = -2.5)
```

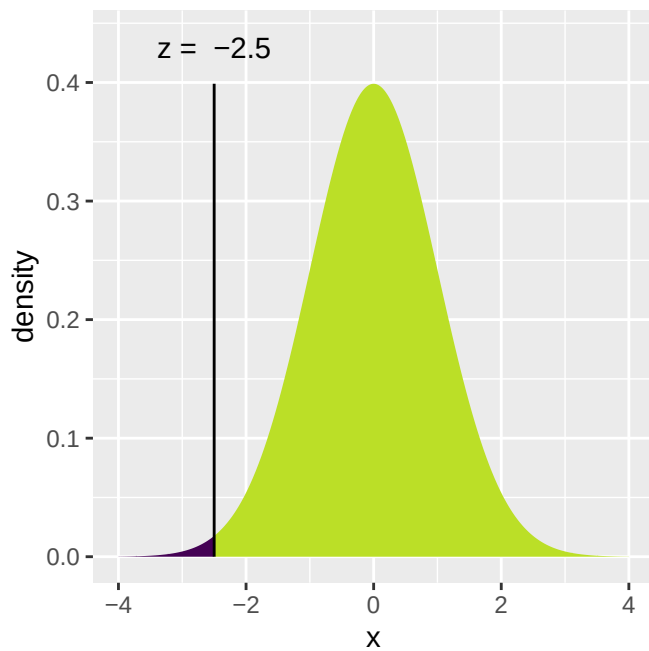
```
#
```

```
#  If X ~ N(0, 1), then
```

```
#   P(X <= -2.5) = P(Z <= -2.5) = 0.00621
```

```
#   P(X >  -2.5) = P(Z >  -2.5) = 0.9938
```

```
#
```




```
#  [1] 0.006209665
```

```r
# b)
mosaic::xpnorm(q = -2.5, mean = -2, sd = 1)
```

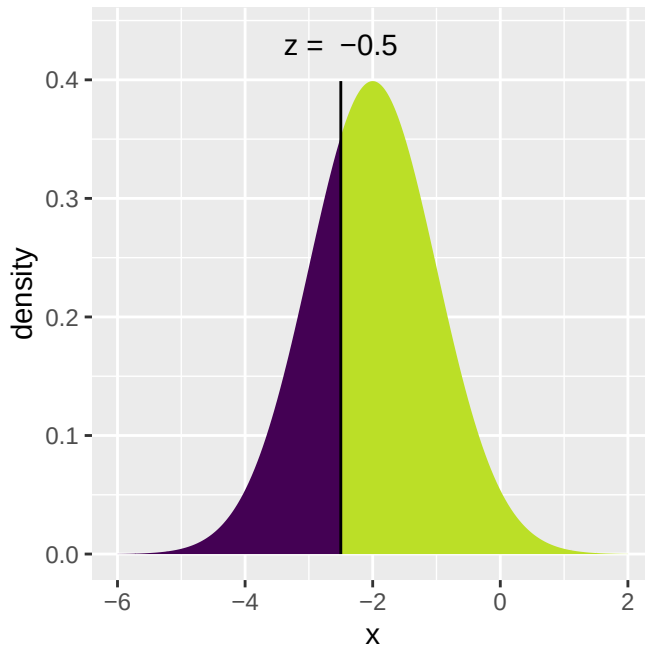
```
#
```

```
#  If X ~ N(-2, 1), then
```

```
#   P(X <= -2.5) = P(Z <= -0.5) = 0.3085
```

```
#   P(X >  -2.5) = P(Z >  -0.5) = 0.6915
```

```
#
```




```
#  [1] 0.3085375
```

```
# c)
mosaic::xpnorm(q = c(-2.5,-1))
```

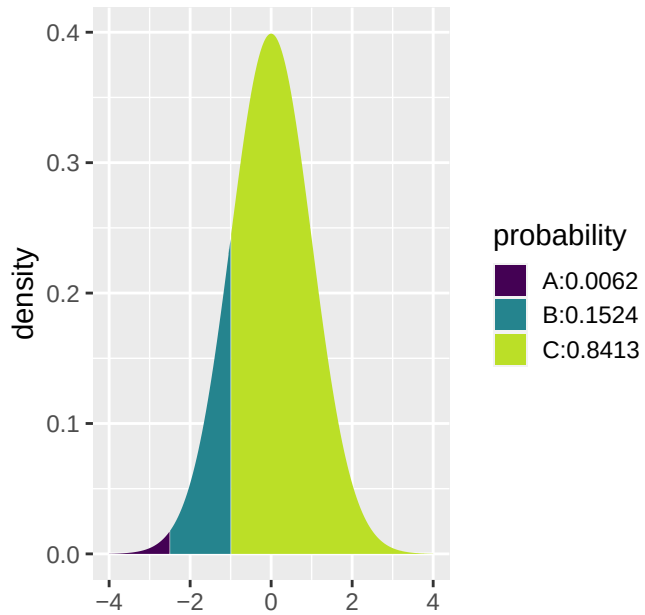
```
#
```

```
#  If X ~ N(0, 1), then
```

```
#   P(X <= -2.5) = P(Z <= -2.5) = 0.00621   P(X <= -1.0) = P(Z <= -1.0) = 0.15866
```

```
#   P(X >  -2.5) = P(Z >  -2.5) = 0.9938    P(X >  -1.0) = P(Z >  -1.0) = 0.8413
```

```
#
```




```
#  [1] 0.006209665 0.158655254
```

```
pnorm(q = -1) - pnorm(q = -2.5)
```

```
#  [1] 0.1524456
```

```
# d)
pnorm(q = -1, mean = -2) - pnorm(q = -2.5, mean = -2)
```

```
#  [1] 0.5328072
```